

Research Paper

A comparative study of enhanced thermal performance in Tesla-type microchannels

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ABSTRACT

Enhancing convective heat transfer in microchannels is significant for cooling of micro-electro-mechanical systems (MEMS). The dominant laminar flow in confined domain at microscale normally results in thick thermal boundary layers, seriously deteriorate the heat transfer process because of limited disturbance of flow. In this study, a novel Tesla-type microchannel configuration with significant flow diodicity was proposed to significantly interrupt boundary layers and promote fluid mixing. Hydrodynamic and thermal performances of this as-designed configuration were comprehensively investigated through experimental and numerical studies. Thermal characteristics such as Nusselt number (Nu), thermal resistance (R_{th}), and performance evaluation criterion (PEC) were computed. Diodic performance was also evaluated by measuring pressure drop diodicity and thermal diodicity in different flow directions. The results demonstrate that convection heat transfer is significantly enhanced because of strong mixing and periodic interruption of liquid flow and thermal boundary layers. Compared to conventional plain-wall microchannel, the average wall temperature is dramatically reduced by about 10 K at $Re = 250$ because of enhanced mixing effect. Accordingly, R_{th} is significantly decreased by $\sim 45\%$. A PEC of 1.3 is achieved as well. Overall, the Nu is ranging from 10 to 25.2 for Re ranging 100 to 450, about 1.3 ~ 2.1 times higher than that of plain-wall microchannel. Diodic performances in terms of Di_p and Di_t are about 1.9 and 1.3 for $D_h = 0.4$ mm. The new understandings of this study might provide insights for the design of new heat sinks for potential applications in electronics cooling.

1. Introduction

The increasing miniaturization of Micro-Electro-Mechanical-Systems (MEMS) usually results in significant rise of power density, posing a great challenge to the cooling of thermal components. Therefore, efficient thermal management plays a crucial role to the safety of electronic devices [1–3]. As a promising cooling technology with effective heat dissipation capability, microchannel heat sinks (MCHSs) have been widely investigated. However, in conventional MCHSs with multiple parallel channels, the long-lasting laminar flow and thick flow and thermal boundary layers significantly impede the enhancement of thermal performance [4,5]. To address this dilemma, various active and passive techniques to interrupt the thermal boundary have been employed. However, active techniques such as magnetic field [6], electric field [7] and mechanical pump [8] need additional pumping power, while passive techniques are cost-effective. Typically, convective heat transfer can be significantly enhanced by replacing working fluid

[9], modifying microchannel surface properties [10] and designing functional geometric constructions [11,12], etc. For example, Al_2O_3 -water and TiO_2 -water nanofluids [9] have been demonstrated to significantly enhance Nu by increasing the thermal conductivity to compensate the low heat transfer rate inside the thermal boundary layer. Furthermore, microporous metal surfaces [10] can dissipate high heat fluxes at lower wall superheat by increasing the heat transfer area. However, the operational durability and complex fabrication of nanofluids and porous surfaces usually limit their practical applications. To date, sufficient interruption of thermal boundary layer and fluid mixing in microchannels still remain.

By now, various geometries and configurations have been exploited and comprehensively investigated to enhance convective heat transfer by disturbing thermal boundary layer and promoting fluid mixing, such as saw-tooth microchannel [13], leaf-vein-shaped microchannel [14,15], wavy microchannel [16] and manifold microchannel [17,18]. Periodic expansion-constriction of cross-sections and variation of flow directions can result in fluid perturbation and redevelopment of thermal

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Nomenclature			
A_c	Convective heat transfer area, m^2	$R_{th,out}$	Outlet thermal resistance, $K/(cm^2 \cdot W)$
AR	Aspect ratio	t	Substrate thickness, m
C_p	Specific heat capacity at 1 atm, $J/(kg \cdot K)$	$T_{m,i}$	Thermocouple measured temperature, K
D_h	Hydraulic diameter, m	$T_{f,ave}$	Average fluid temperature, K
Di_p	Pressure drop diodicity	$T_{f,in}$	Inlet fluid temperature, K
Di_t	Thermal diodicity	$T_{w,ave}$	Average wall temperature, K
f	Average apparent friction factor	$T_{w,max}$	Maximum wall temperature, K
$f_{app,ave}$	Apparent friction factor by Shah and London	$T_{w,out}$	Wall temperature at outlet, K
h_{ave}	Heat transfer coefficient, $W/(m^2 \cdot K)$	$T_{w,i}$	Local wall temperature, K
H_{ch}	Unit microchannel height, m	ΔT	Temperature difference, K
I	Electrical current, A	ΔT_{in-out}	Temperature difference between inlet and outlet, K
k_s	Copper thermal conductivity, $W/(m \cdot K)$	u	Fluid velocity, m/s
k_f	Fluid thermal conductivity, $W/(m \cdot K)$	u_m	Average inlet fluid velocity, m/s
L	Length of the channel, m	V	Voltage, V
L_t	Unit Tesla-type bend structure length, m	W	Width of microchannel, m
Nu	Nusselt number	W_{ch}	Unit channel width, m
P	Pressure, Pa	W_d	Narrow section distance, m
P_{in}	Inlet pressure, Pa	W_t	Fin thickness, m
P_{out}	Outlet pressure, Pa	Greek symbols	
ΔP	Pressure drop, Pa	ρ	Density, kg/m^3
ΔP_B	Backward pressure drop, Pa	Γ	Interface
ΔP_c	Abrupt contraction pressure drop at inlet, Pa	μ	Dynamic viscosity, Pa·s
ΔP_e	Abrupt expansion pressure drop at outlet, Pa	Φ	Viscous dissipation function
ΔP_F	Forward pressure drop, Pa	Subscripts	
PEC	Performance evaluation criterion	<i>ave</i>	Average
P_{in}	Inlet pressure, Pa	<i>B</i>	Backward
P_{out}	Outlet pressure, Pa	<i>eff</i>	Effective
Po	Poiseuille number	<i>f</i>	Fluid
q_{eff}	Effective heat flux, W/cm^2	<i>F</i>	Forward
Q_{eff}	Effective heating power, W	<i>h</i>	Hydraulic
Q_{in}	Inlet flow rate, ml/min	<i>i</i>	Thermocouple location
Q_{loss}	Heat loss, W	<i>in</i>	Inlet
Q_{total}	Total input power, W	<i>out</i>	Outlet
R_d	Radius of unit bend, m	<i>s</i>	Solid
Re	Reynold number	<i>w</i>	Wall
R_{th}	Thermal resistance, K/W	<i>O</i>	Base plain microchannel

boundary layers. For example, strong vortices were induced in the saw-tooth microchannels, leading to a significant enhancement of Nu up to 50 % [13]. Leaf-vein-shaped microchannels [19] can result in decrease of maximum temperature difference between inlet and outlet up to 66 % and the COP is two-fold than that in conventional serpentine channels. By optimizing pin fins arrangement with a ratio of transverse spacing to longitudinal spacing ranging 0.49 to 1.34, manifold microchannels [17] exhibited substantially enhanced convective heat transfer performance. These aforementioned studies demonstrated that convective heat transfer can be considerably increased by generating strong fluid mixing and interrupting thermal boundary layers.

Intensified fluid mixing can also be achieved by introducing additively ribs, fins, baffles and other various vortex generators, etc. Typically, Nu can be increased by inserting multi-layers of twisted porous ribs [20]. Despite significant increase of pressure drop, the manufactured static mixers were able to enhance the heat transfer rate by two-fold compared to that of plain channels [21]. Chevron fins achieved an enhanced thermal performance accompanying with a reduction of thermal resistance of 19.5 % by generating secondary flow [22]. To assess the overall performance of MCHSs, performance evaluation criterion (PEC) is one of most important factors [23]. PEC is determined by the values of Nusselt number (Nu) and friction factor (f). The merit of new design can be illustrated by PEC values greater than 1. For example, microchannels with additional cavities [5] achieved a PEC of 1.62 owing

to chaotic mixing. In contrast, cylindrical vortex generators [24] resulted in a PEC approximately 0.8, due to limited enhancement of thermal performance and a substantial increase of pressure drop.

Strong and chaotic fluid mixing can be generated by Tesla valves structures in microchannels [12,25,26]. This effectiveness can be further enhanced through an in-series arrangement with a number of stages $N < 10$ [25]. The results indicated that Tesla valves [26] could enhance momentum exchange by altering the flow direction, leading to a significant increase of Nu about 111.2 %. Tesla valves also exhibit strong flow diodicity [2,12]. Nevertheless, the influence of Tesla-type structures on single-phase heat transfer performances have not been extensively investigated. In our previous work [27], we have developed a novel Tesla-type microchannel and conducted experimental studies. This new configuration leads to a significant enhancement of flow boiling performance by effectively inhibiting vapor backflow and enhancing two-phase mixing. In this study, comprehensively experimental and numerical studies will be conducted to investigate the effect of this Tesla-type microchannel on the single-phase heat transfer.

The overall performance of this new configuration will be quantitatively assessed by thermal and flow characteristics such as Nu , $T_{w,ave}$, f , R_{th} and PEC, etc. Also, the diodic performance of this design will be discussed through measuring thermal performance and pressure drop in different flow directions. Finally, the effect of hydraulic diameter (D_h) on overall performance will be studied as well. Firstly, the actual MCHS

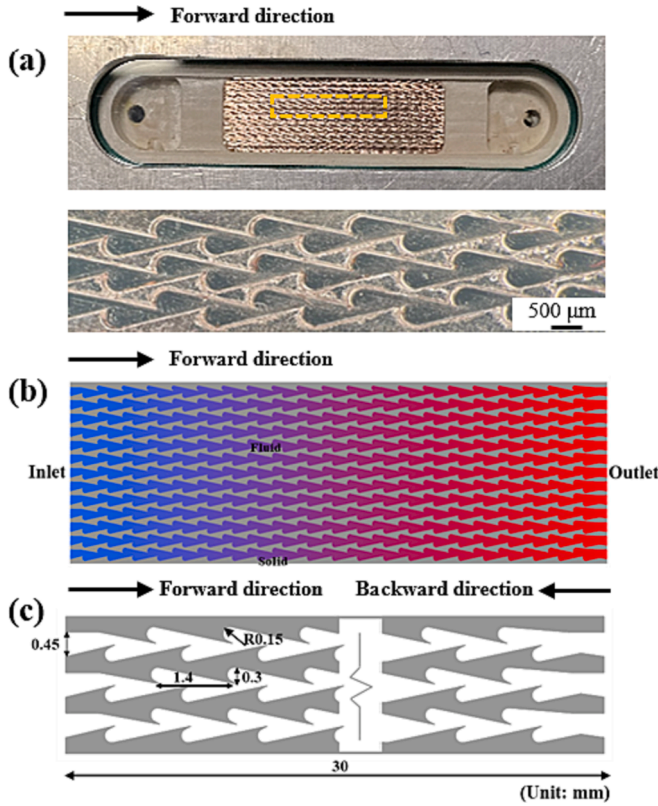


Fig. 1. Tesla-type microchannel configuration (a) manufactured microchannels, (b and c) Schematic of microchannels.

design, testing system and experimental procedures are described in section 2. Secondly, section 3 establishes the numerical model. The characterizations of hydrodynamic and thermal performance are illustrated in section 4. Finally, we present a comprehensive experimental and numerical results in section 5 and discuss the enhanced mechanism of convective heat transfer.

2. Channel design and experimental measurements

2.1. Rational design of microchannel

To enhance fluid mixing in the confined domain, a Tesla-type microchannel was designed. Fig. 1 shows schematic and actual image of Tesla-type microchannel configuration based on micro-milling process. Because of high thermal conductivity and low-cost fabrication, copper was selected as microchannel material. As shown in Fig. 1 (c), the width (W_{ch}) and height (H_{ch}) of each parallel microchannel are 0.45 mm

and 0.5 mm, respectively. Fin thickness (W_f) between two parallel microchannels was 0.22 mm. Length (L_f) of unit bend was 1.4 mm and radius (R_d) of bend was 0.15 mm.

Fig. 2 illustrates the schematic cross-section for this Tesla-type microchannel heat sink. To prevent water leakage, a silicon O-ring was placed between the microchannel and quartz glass. Copper microchannel was packed with ceramic fiber insulating layer to minimize heat loss. To obtain wall temperature, seven T-type thermocouples were uniformly embedded at a distance of 2.5 mm below the substrate. The arrangement of these thermocouples formed a “T” shape, with horizontal and vertical distances of 6.5 mm and 2.5 mm between adjacent thermocouples.

2.2. Experimental system

Fig. 3 illustrates the detail experimental set-up. DI water from a 30-liter water reservoir was driven by the high pressure compressed nitrogen. After passing through a 8 μ m filter, flow rates were monitored and regulated by gear flowmeter (accuracy 0.5 %) and high-precision valve. To obtain inlet and outlet temperature, two T-type thermocouples (Omega, accuracy ± 0.4 °C) were placed inside plenums. Pressure drop was measured through two pressure transducers (PCM300D, accuracy 0.3 %). Inlet fluid temperature was controlled through the band preheater. Heating power was supplied through a digital power. Main parameters including temperature and pressure were recorded by using an Agilent 34970A data acquisition device. Fig. 3b presents an actual image of experimental set-up.

2.3. Experimental measurements

Before experiments, heat loss calibration was conducted by heating copper microchannel without fluid. Detailed testing process of heat loss calibration can refer to our previous researches [27,28]. The correlation between heat loss and average wall temperature can be obtained by:

$$Q_{loss} = 0.2656T_{w,ave} - 7.98 \quad (1)$$

In this work, inlet fluid temperature was kept constantly at about 30 °C. Parameters like voltage, current, temperature, pressure and flow rates were collected to evaluate the hydrodynamic and thermal performances. Effective power is calculated by:

$$Q_{eff} = Q_{total} - Q_{loss} \quad (2)$$

Where input power Q_{total} can be obtained by current multiplying voltage, heat loss Q_{loss} is defined by Eq.(1). Based on one-dimensional Fourier law, local substrate temperature can be calculated by:

$$T_{w,i} = T_{m,i} - \frac{q_{eff}t}{k_s} \quad (i = 1, 2, 3, 4, 5) \quad (3)$$

Where $T_{m,i}$ is the measured local temperature, k_s is copper thermal

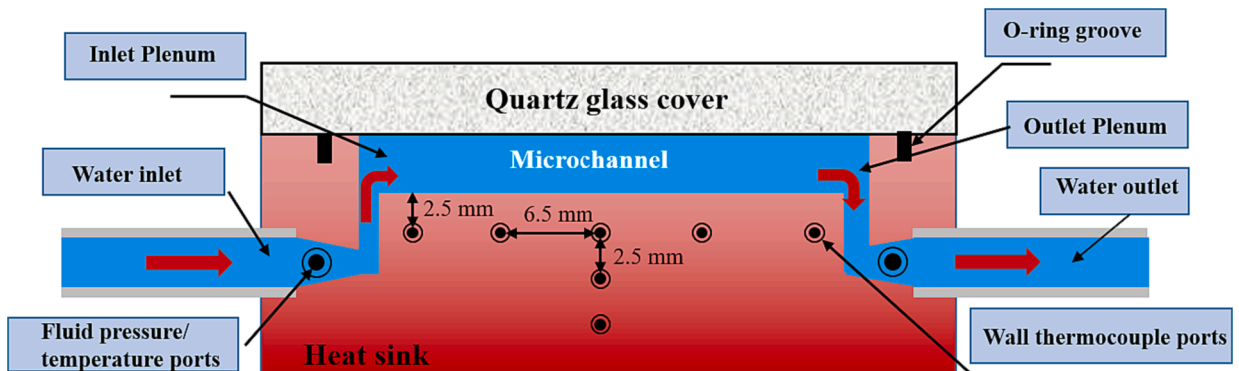


Fig. 2. Schematic cross-section of Tesla-type microchannel heat sink.

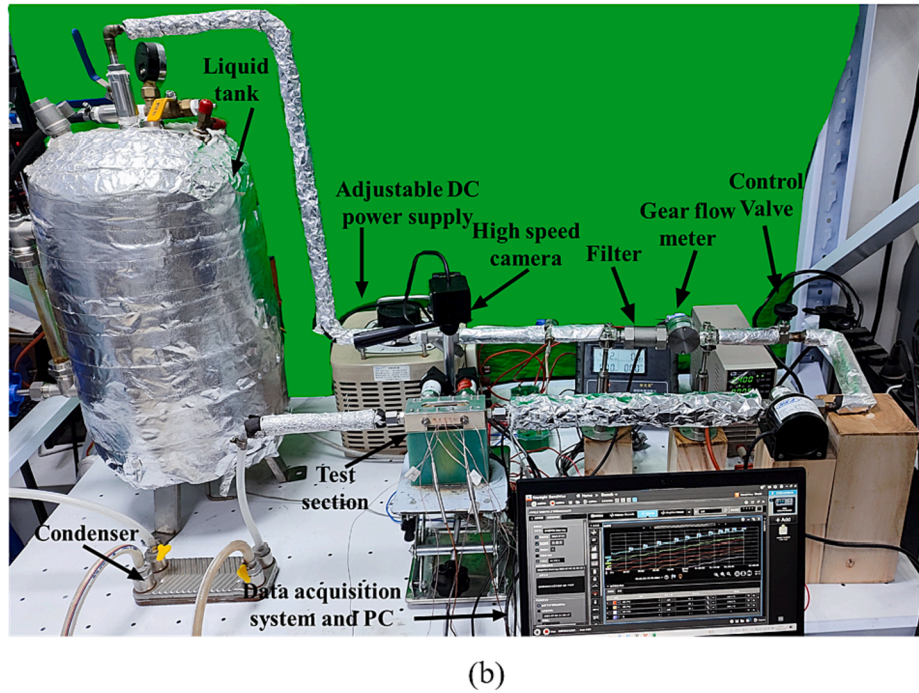
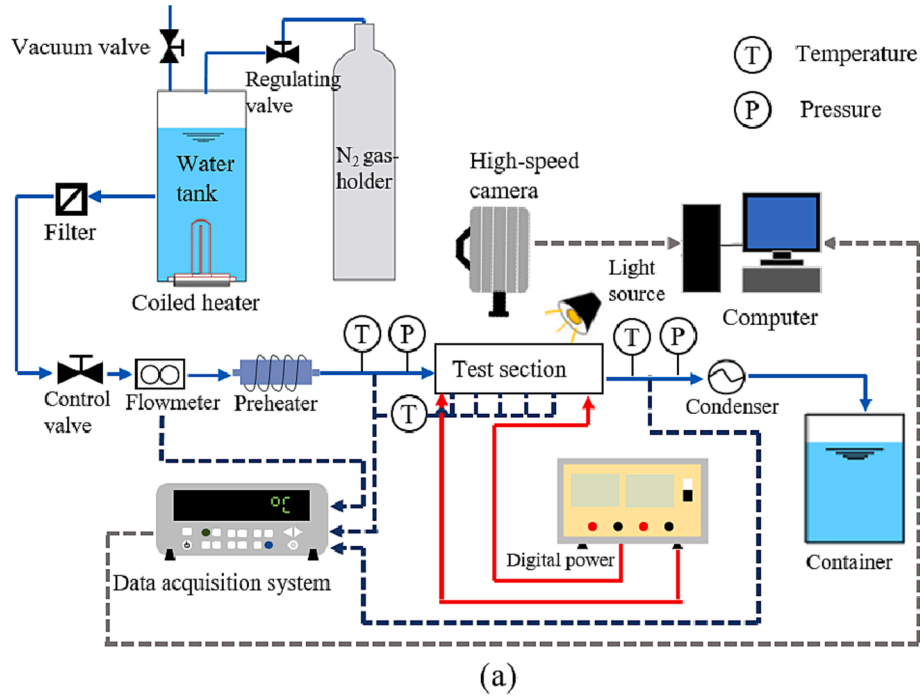


Fig. 3. (a) Schematic drawing and (b) actual image of experimental flow loop.

conductivity. t represents vertical distance between the thermocouple and microchannel substrate, about 2.5 mm. Therefore, average temperature ($T_{w,ave}$) of microchannel substrate is expressed by:

$$T_{w,ave} = \frac{T_{w,1} + T_{w,2} + T_{w,3} + T_{w,4} + T_{w,5}}{5} \quad (4)$$

Single-phase pressure drop is evaluated by:

$$\Delta P = (P_{in} - \Delta P_c) - (P_{out} + \Delta P_e) \quad (5)$$

Where ΔP_c and ΔP_e are abrupt contraction pressure drop and expansion pressure drop. P_{in} and P_{out} are measured pressures at inlet and outlet. According to existed studies [29,30], the ΔP_c and ΔP_e are negligible, less than 5 % of total measured pressure drop. Uncertainties

of main parameters such as temperature, flow rate and pressure are about ± 0.4 °C, ± 0.5 % and ± 0.3 %, respectively. Based on error propagation method [31], the uncertainties of Q_{eff} and Nu are 4.2 % and 8.9 %, respectively. More details about the data deduction are provided in the previous studies [27,28].

3. Numerical method

3.1. Physical model

A copper microchannel heat sink consisting of multiple parallel Tesla-type microchannels was designed in this work (See Fig. 4(a)). Main dimensions in this model are consistent with the experimental

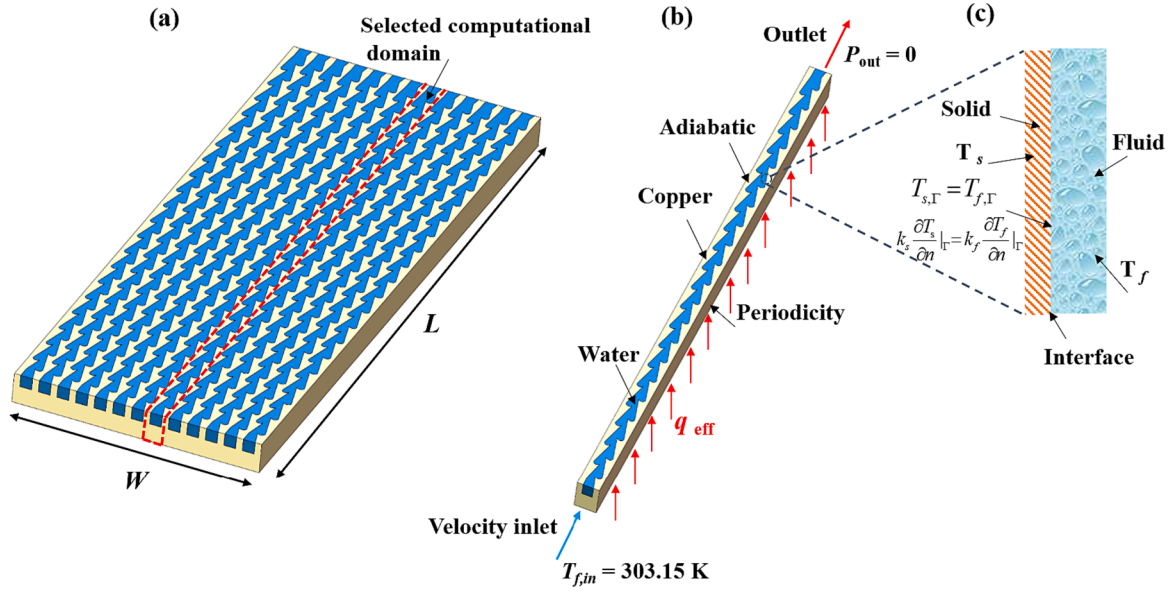


Fig. 4. Schematic of Tesla-type microchannel configuration and boundary conditions.

Table 1

Geometry dimensions of Tesla-type microchannel configuration.

Parameters	Unit	Values
Heatsink length (L)	mm	30
Heatsink width (W)	mm	10
Channel width (W_{ch})	mm	0.45
Channel height (H_{ch})	mm	0.5
Fin thickness (W_f)	mm	0.22
Tesla-type bend structure length (L_t)	mm	1.4
Radius of Tesla bend (R_d)	mm	0.15
Narrow section width (W_d)	mm	0.22

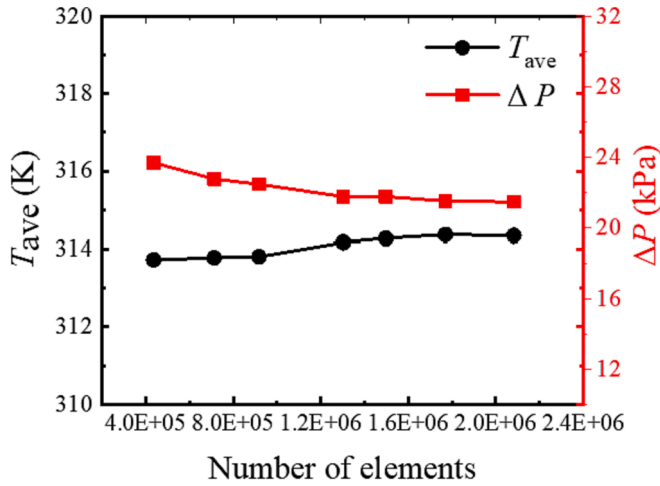


Fig. 5. Mesh independence study.

study. Table 1 shows the dimensions of this Tesla-type microchannel configuration. In consideration of symmetry, a computational domain was established by using one single Tesla-type microchannel.

3.2. Governing equations

To successfully characterize hydrodynamic and thermal performances of Tesla-type microchannel configuration, several simplifying

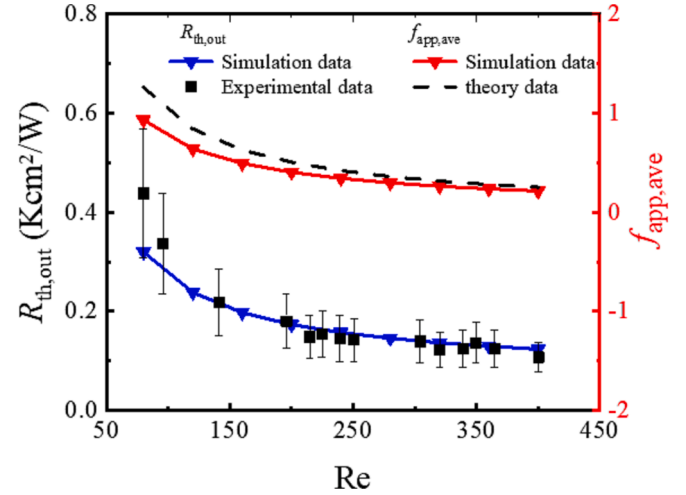


Fig. 6. A comparative analysis of current model against theoretical friction factor [37] and experimental outlet thermal resistance [39].

assumptions could be made as:

- (1) Laminar and incompressible Newtonian fluid;
- (2) Gravity effect and other body forces are ignored;
- (3) Physical properties of solid microchannel are constant; Fluid properties are temperature dependent;
- (4) The fluid is in single-phase state without phase change.

Governing equations for this physical model are [5,32]:

$$\nabla \cdot \vec{u} = 0 \quad (6)$$

$$\rho \vec{u} \cdot \nabla \vec{u} = -\nabla P + \mu \nabla^2 \vec{u} \quad (7)$$

$$\rho c_p (\vec{u} \cdot \nabla T_f) = k_f \nabla^2 T_f + \Phi \quad (8)$$

Where u , c_p , μ are fluid velocity, specific heat and viscosity, respectively. P and ρ are pressure and liquid density. T_f and k_f are fluid temperature and thermal conductivity. Φ is the viscous dissipation generated by viscous friction, however, based on previous studies

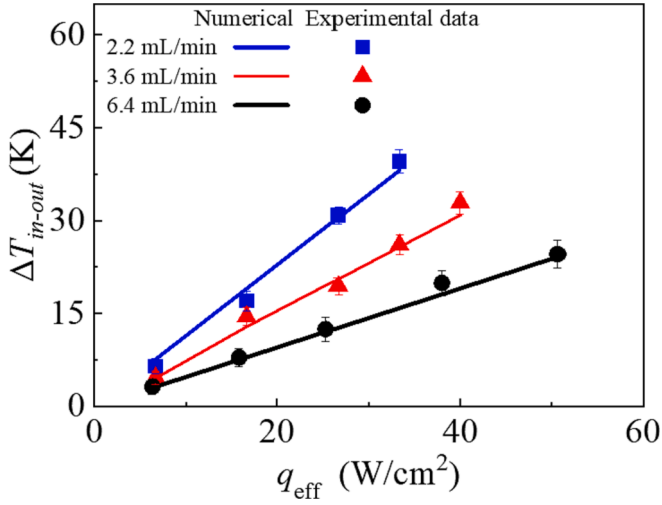


Fig. 7. A comparative analysis between numerical and experimental studies.

[33–35], viscous dissipation would be ignored due to large micro-channel structure size and low viscosity of fluid.

Energy equation for solid domain:

$$k_s \nabla^2 T_s = 0 \quad (9)$$

Thermo-physical parameters of solid are constant. Fluid properties are functions of temperature [24], which are defined as:

$$\begin{aligned} \rho(T_f) &= 838.466135 + 1.40050603T_f - 0.0030112376T_f^2 \\ &\quad + 3.71822313 \times 10^{-7}T_f^3 \\ \mu(T_f) &= 1.3799566804 - 0.021224019151T_f + 1.3604562827 \times 10^{-4}T_f^2 \\ &\quad - 4.6454090319 \times 10^{-7}T_f^3 + 8.9042735735 \times 10^{-10}T_f^4 \\ &\quad - 9.0790692686 \times 10^{-13}T_f^5 + 3.8457331488 \times 10^{-16}T_f^6 \\ c_p(T_f) &= 12010.1471 - 80.4072879T_f + 0.309866854T_f^2 \\ &\quad - 5.38186884 \times 10^{-4}T_f^3 + 3.62536437 \times 10^{-7}T_f^4 \\ k_f(T_f) &= -0.869083936 + 0.00894880345T_f \\ &\quad - 1.58366345 \times 10^{-5}T_f^2 + 7.97543259 \times 10^{-9}T_f^3 \end{aligned} \quad (10)$$

3.3. Boundary conditions

Settings of boundary conditions are illustrated in Fig. 4(b). Inlet liquid temperature was set to be 303.15 K. Inlet liquid velocity is set based on Re , which ranges from 50 to 400 in this study. A constant heat flux of $Q_{eff} = 16.5 \text{ Wcm}^{-2}$ is applied at bottom substrate and other surfaces are adiabatic. At solid walls, no-slip velocity of $u_s = 0$ is assigned. For solid–liquid interfaces, heat transfer is coupled by heat flux

continuity, where $T_{s,\Gamma} = T_{f,\Gamma}$ and $k_s \frac{\partial T_s}{\partial n}|_{\Gamma} = k_f \frac{\partial T_f}{\partial n}|_{\Gamma}$, Γ represents interface between water and copper.

4. Data reduction

To characterize overall performance of this Tesla-type microchannel, various variables were defined and calculated in this work [23,25,36], such as average heat transfer coefficient (h_{ave}), thermal resistance (R_{th}), average apparent friction factor (f), Nu , PEC , Di_p and Di_t .

The Reynold number (Re) is expressed by:

$$Re = \frac{\rho u_m D_h}{\mu} \quad (11)$$

Where hydraulic diameter D_h is defined by:

$$D_h = \frac{2H_{ch}W_{ch}}{(H_{ch} + W_{ch})} \quad (12)$$

In the present Tesla-type microchannel, characteristic width equals to mean value of minimum and maximum width of cross-section.

Thermal performance is evaluated by Nu , which can be obtained by:

$$h_{ave} = \frac{Q_{eff}}{A_c (T_{w,ave} - T_{f,ave})} \quad (13)$$

$$Nu = \frac{h_{ave} D_h}{k_f} \quad (14)$$

where A_c is convective heat transfer area. $T_{w,ave}$ and $T_{f,ave}$ represent average copper and water temperature.

Average apparent friction factor can be given by:

$$f = \frac{\Delta P D_h}{2\rho L u_m^2} \quad (15)$$

For laminar flow in the plain microchannel, the apparent friction factor is compared with results by Shah and London [37], which can be expressed by:

$$f_{app,ave} Re = \sqrt{\left[\frac{3.2}{(L/ReD_h)} \right]^2 + (Po)^2} \quad (16)$$

The Poiseuille number (Po) can be given as:

$$Po = 96 \left[1 - 1.3553(AR) + 1.9467(AR)^2 - 1.7012(AR)^3 + 0.9564(AR)^4 - 0.2537(AR)^5 \right] \quad (17)$$

Where AR represents microchannel aspect ratio.

Thermal resistance (R_{th}) is expressed by [38]:

$$R_{th} = \frac{\Delta T}{Q_{eff}} = \frac{T_{w,max} - T_{f,in}}{Q_{eff}} \quad (18)$$

Where $T_{w,max}$ and $T_{f,in}$ are respectively maximum temperature of solid wall and inlet fluid temperature.

The PEC [23,36] is used to show the merits of as-designed Tesla-type

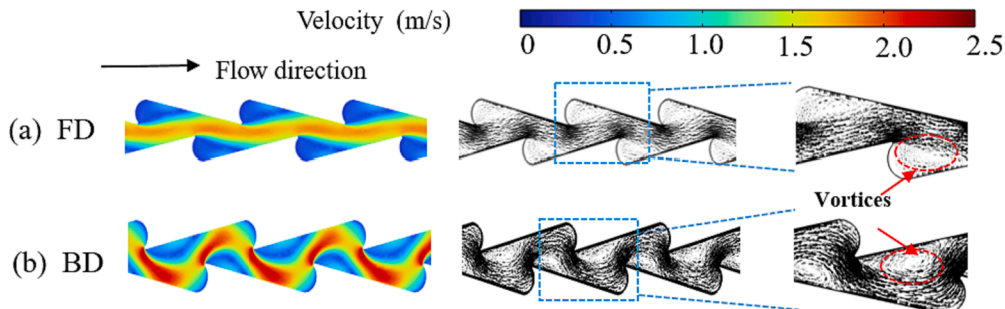


Fig. 8. Flow fields and streamlines in two different flow directions.

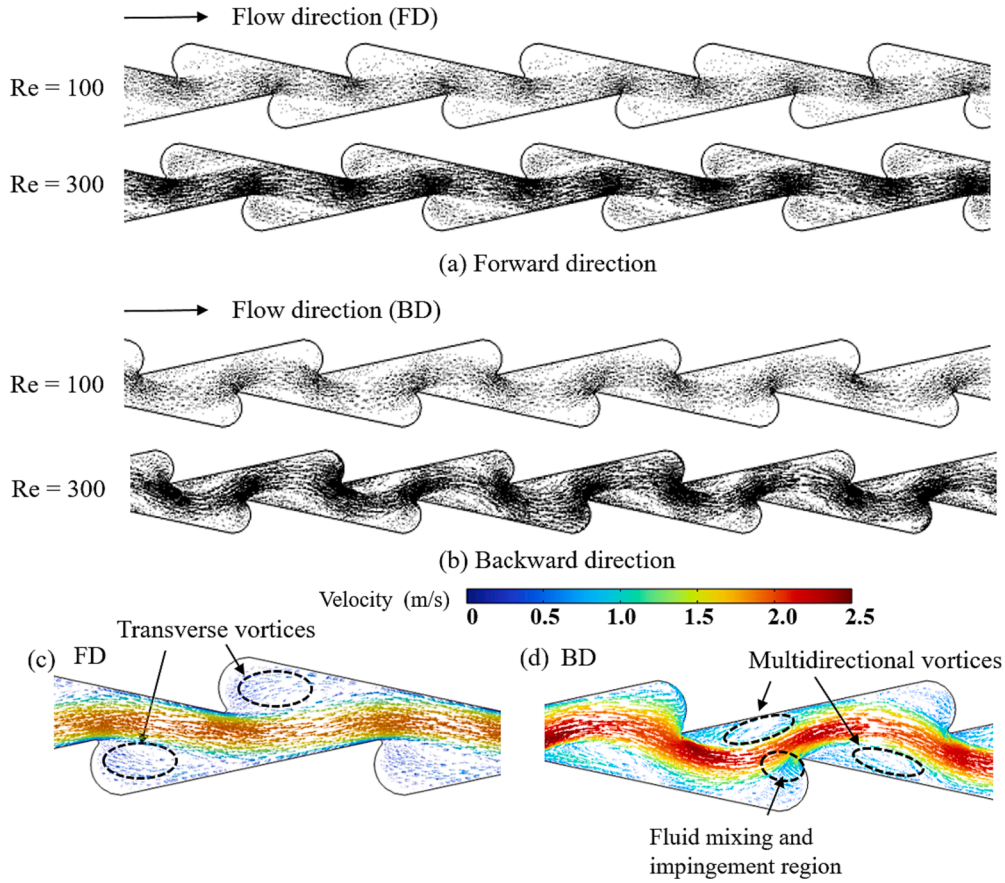


Fig. 9. Enhanced fluid mixing in two different flow directions.

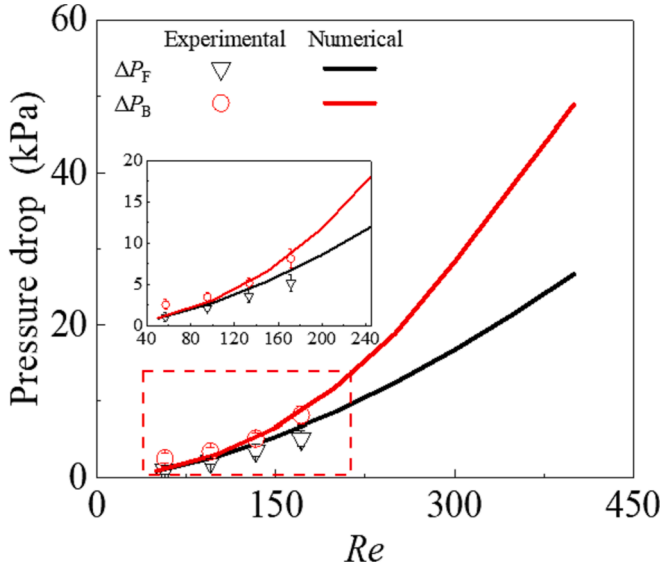


Fig. 10. Pressure drop variation in different flow directions.

microchannel with the same effective power consumption.

$$PEC = \frac{Nu/Nu_0}{(f/f_0)^{1/3}} \quad (19)$$

The subscript 0 denotes the conventional plain microchannel.

To quantitatively characterize the diodic performance of this Tesla-type microchannel, pressure drop diodicity (Di_p) and thermal diodicity

(Di_t) were calculated, which were respectively defined as the ratio of pressure drop (ΔP) and Nusselt number (Nu) in the backward and forward direction [25]:

$$Di_p = \frac{\Delta P_B}{\Delta P_F} \quad (20)$$

$$Di_t = \frac{Nu_B}{Nu_F} \quad (21)$$

Subscript B and F denote backward and forward direction, respectively.

5. Results and discussion

5.1. Mesh independence study and model validation

Governing equations were solved by using finite element method (FEM) with COMSOL Multiphysics software. In this study, tetrahedral-type unstructured meshes were selected for complex geometries. The meshes were refined in the boundary layers near solid walls. The relative error tolerance was set to be $1e-5$ to ensure numerical accuracy. Also, mesh independence study was conducted in this Tesla-type microchannel. Fig. 5 indicates that average wall temperature and pressure drop in the backward direction become convergent as the increasing of element number, ranging from $4.3 \text{ e}5$ to $2.1 \text{ e}6$. Based on this study, the refined meshes with $1.7e6$ element number were selected for the computation of our physical model.

Furthermore, numerical results were evaluated against existing experimental studies to confirm the reliability of current numerical study [37,39]. In the Fig. 6, we compare numerical models with theoretical average friction factor results presented by Shah and London [37]

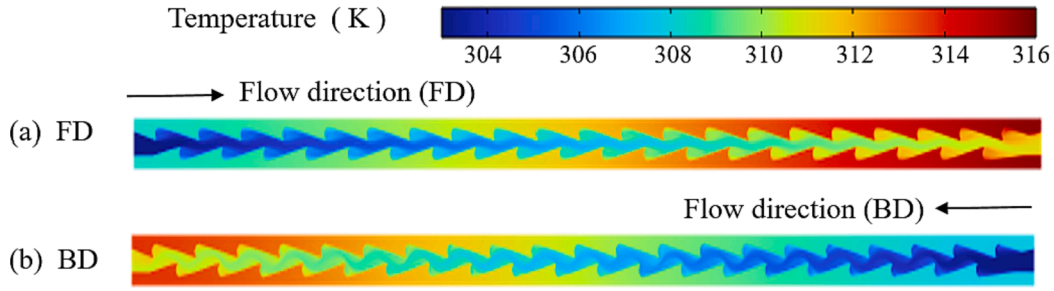
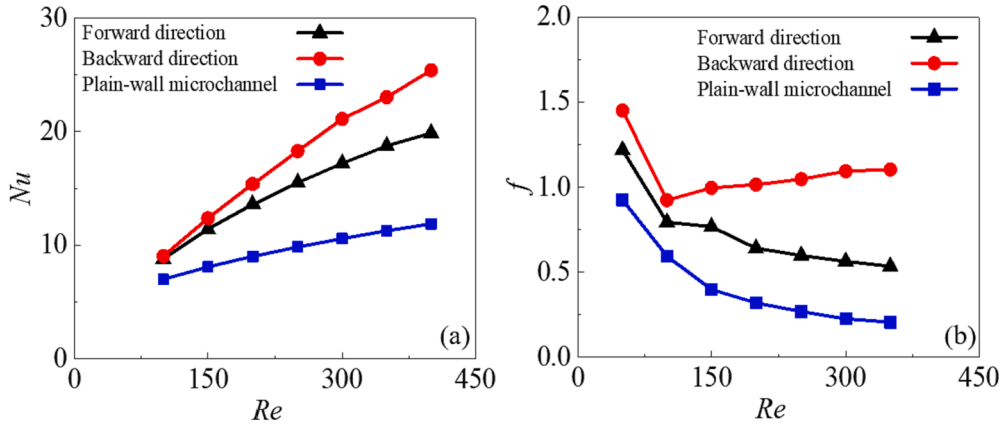
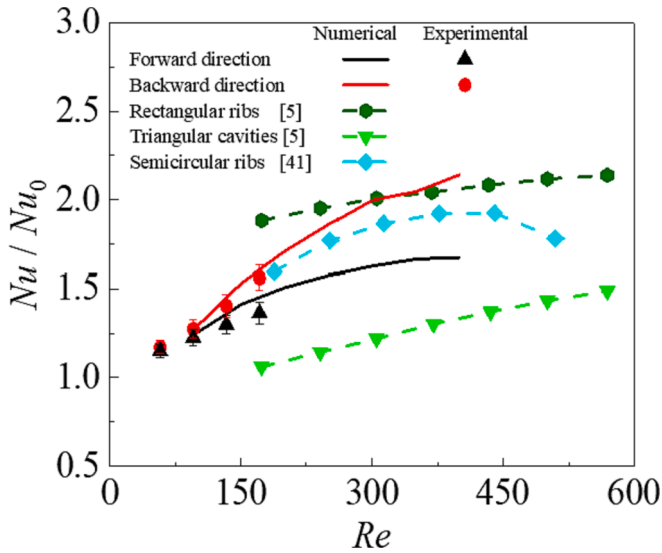


Fig. 11. Wall temperature distribution along different flow directions.

Fig. 12. Comparison of Nu (a) and friction factor (b) among different microchannel configurations.Fig. 13. Nu/Nu_0 against Re in the two different directions.

and experimental outlet thermal resistance data provided by Qu and Mudawar [39]. For single-phase heat transfer, upper surface of silicon microchannel applied constant heat flux ($q_{eff} = 90 \text{ W/cm}^2$) and inlet water temperature maintains 293.15 K. The size (length $\times$ width $\times$ height) of silicon microchannel is $10 \text{ mm} \times 57 \mu\text{m} \times 180 \mu\text{m}$. The outlet thermal resistance ($R_{th,out} = (T_{w,out} - T_{f,in}) / q_{eff}$) is assessed for Re ranging from 80 to 100. Besides, the theoretical friction factor is computed according to Eq. (16). Fig. 6 indicates this numerical model nicely agree with experimental and theoretical results with a maximum relative error of about 9.5 %.

For single-phase experimental study, various parameters including

current, voltage, inlet and outlet fluid temperature will be measured at different flow rates. In the numerical study, the settings of boundary conditions are consistent with the experimental studies for Tesla-type microchannel. Consequently, Fig. 7 shows the numerical results also match with our experimental data very well with an average relative error of only 6.4 %.

5.2. Fluid flow characteristics

This Tesla-type microchannel configuration can significantly enhance fluid flow performance. To demonstrate enhanced effect, a comprehensive analysis of flow field was conducted. Fig. 8 illustrates flow distribution in the x-y plane ($z = 0.25 \text{ mm}$) at $Re = 300$ in the different flow directions. Periodic Tesla-type bends leads to a distinct “S” flow path compared to plain microchannel, where the streamlines are parallel without mixing [5]. The streamlines in our Tesla-type microchannels are characterized “S” shape, indicating the fluid is strongly mixed owing to the perturbation of Tesla-type bends. In contrast to the forward direction, the fluid mixing is significantly enhanced in backward direction with periodic sharp tips of Tesla-type structures. Previously, hydrodynamic performance of multistage Tesla valves were investigated [25,40]. Their results demonstrated Tesla valves have effect on fluid mixing.

Fig. 9 further shows the Re is vital to the fluid mixing in this Tesla-type microchannels. Fig. 9 compares the velocity vectors between forward and backward direction at Re of 100 and 300, respectively. Flow boundary layers are disturbed accompanying with transverse vortices in forward direction. The intensity of these vortices significantly enhances at higher Re [25,26]. In the backward direction, significant flow bifurcation and fluid mixing is observed according to the flow field regarding the velocity vectors. Therefore, thermal performance would be dramatically enhanced owing to this substantially enhanced fluid mixing.

The introduction of this Tesla-type structure might influence the flow

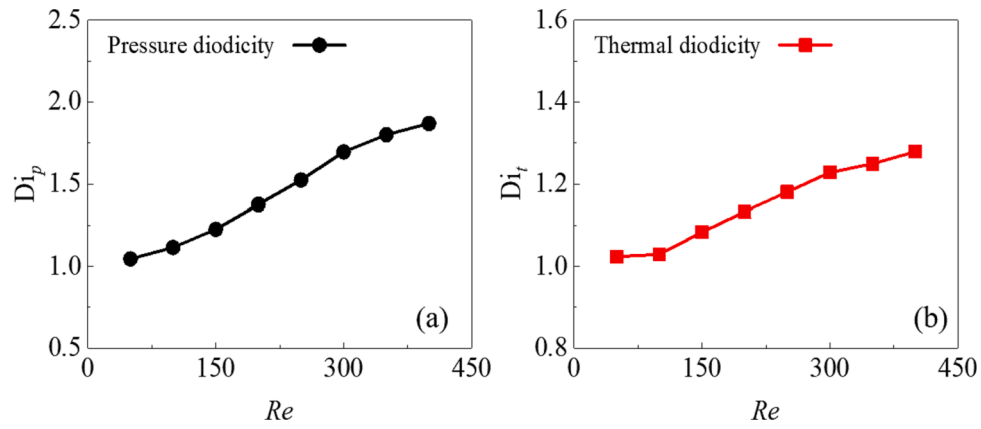


Fig. 14. Diodic performance for Tesla-type microchannels: (a) Pressure diodicity and (b) Thermal diodicity.

Table 2

Comparison of thermal and hydrodynamic performance among different copper microchannel configurations.

Microchannel configurations	Material	Fluid	Enhancement	Enhanced mechanisms
Serpentine microchannel [38]	Copper	DI water	Nu_{avg} increased by 35 %; Pressure drop up to ten-fold; The maximum PEC up to ~ 1.13 .	The channel bends would prevent the fully-developed thermal and hydrodynamic boundaries.
Manifold microchannel [42]	Copper	DI water	Pressure drop reduced by 90 %; Thermal performance index increased by 139 %	Narrowed manifold would equalize pressure and uniformly distribute inlet fluid.
Insertion of twisted tape [43]	Copper	DI water	Friction factor ratio ranged 15.05–17.51	Increasing convective heat transfer area; Generated mixing would induce turbulence.
Micro pin-fin microchannel [44]	Copper	DI water	Enhanced heat transfer achieved at cost of pressure drop	Interrupting and redeveloping boundary layers.
Present study	Copper	DI water	Nu increased by 30 %–110 %; PEC up to 1.3	Strong mixing between central cold water and near-wall hot water; Periodic interruption of thermal and hydrodynamic boundary layers

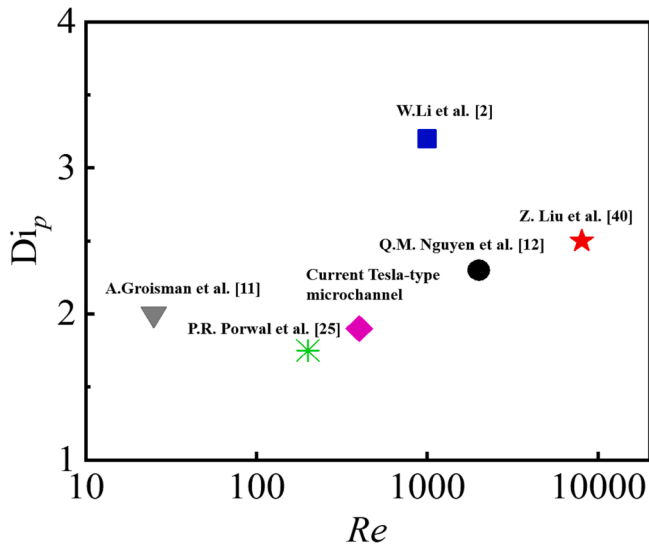


Fig. 15. The pressure diodic performance among different studies.

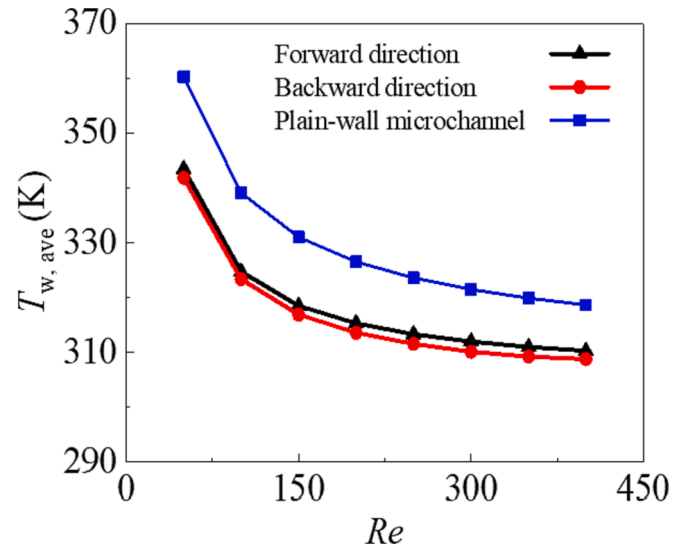


Fig. 16. Variation of average wall temperature against Re.

resistance. Fig. 10 illustrates variation of pressure drop in experimental and numerical studies. Pressure drops gradually increase versus Re in both directions due to the escalating friction forces [21,38]. Due to the inherent flow diodicity of our Tesla-type microchannel, the backward flow resistance is much higher than that in the forward direction.

5.3. Heat transfer characteristics

Sufficient fluid mixing and chaotic advection can significantly increase the thermal performance. In this section, heat transfer

characteristics of this Tesla-type microchannel configuration were numerically studied. Fig. 11 presents wall temperature variation along flow direction. From the inlet to outlet, wall temperature gradually increases owing to continuous conjugate heat transfer process. Enhanced fluid mixing in backward direction leads to lower wall temperature near the outlet section compared to the forward direction. Previous studies demonstrated that the periodic cavities or ribs have significant effect on the thermal performance [5,34].

To demonstrate the increased thermal performance of this design, Fig. 12 (a) plots the significantly increased Nu compared to that of plain

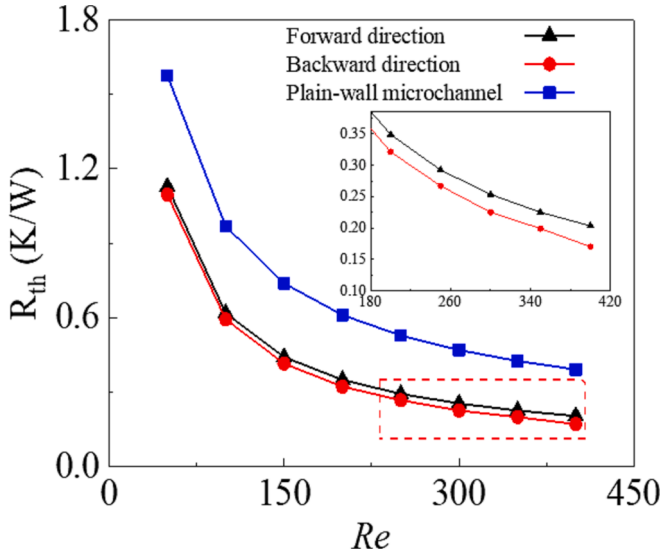


Fig. 17. Thermal resistance against Re.

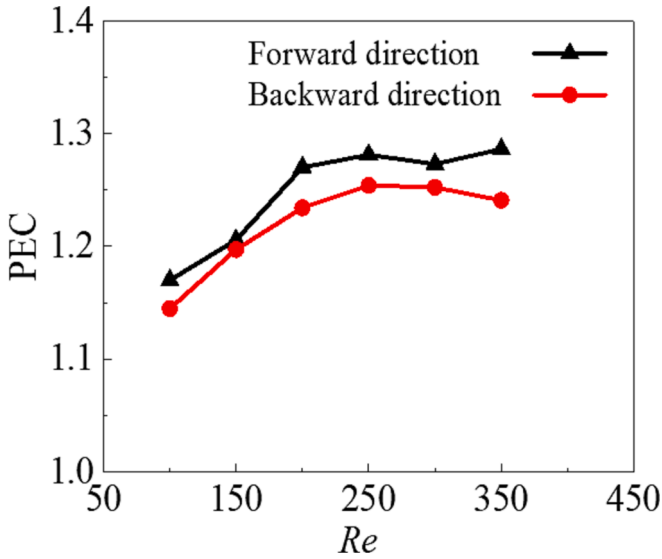
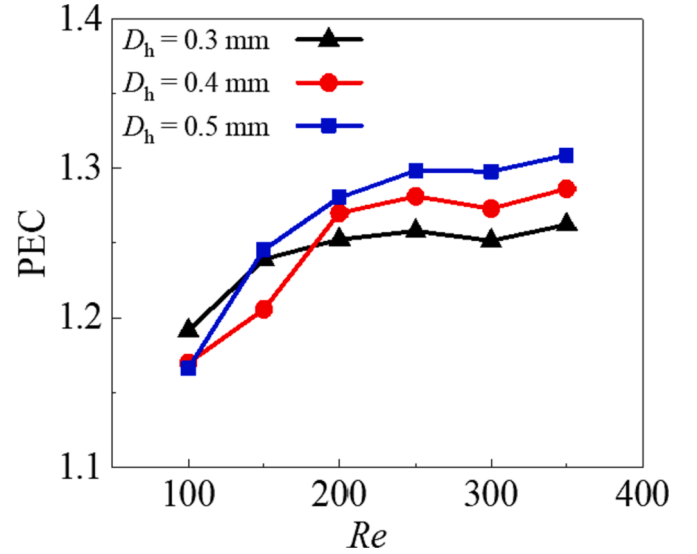
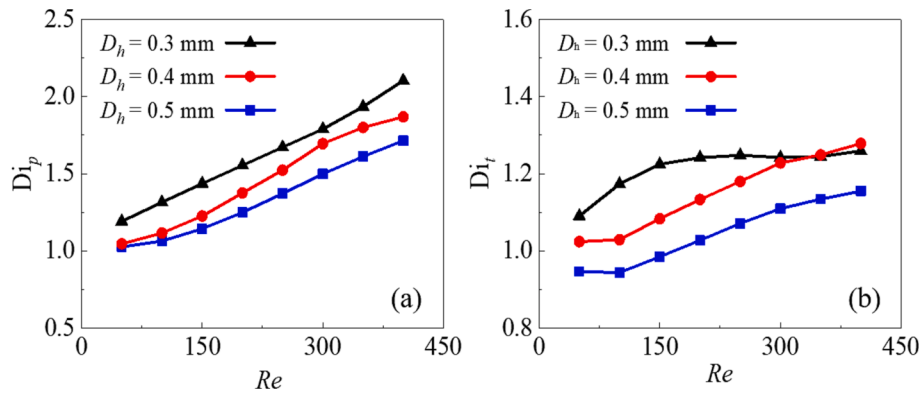


Fig. 18. Variation of PEC in different flow directions.

wall microchannel in both directions. Continuous interruption of thermal boundary layers and sufficient mixing contribute to this enhancement of Nu . Inherently, on the hand, the features of this design would

induce higher flow resistance, particularly in the backward direction. To quantitatively evaluate this effect, Fig. 12 (b) presents the friction factor versus Re . At $Re < 100$, all friction factors show a laminar-like scaling ($f \sim Re^{-1}$). As $Re > 100$, the friction factor of the present transits to turbulent-like scaling ($f \sim Re^0$). This is consistent with reported study by Q.M. Nguyen et al in 2021 [12]. Additionally, in this study, the diodic performance of this design is activated when $Re > 100$ (as shown in Fig. 14). Fig. 13 illustrates the variation of Nu/Nu_0 against Re through experimental and numerical studies between the current study and others' studies [5,41]. Nu_0 refers to the result of conventional plain microchannel configuration. For the existed microchannels with semi-circular ribs [41], rectangular cavities and rectangular ribs [5], the chaotic advection was achieved owing to sudden-expansion flow areas. In this study, periodic Tesla-type bends can induce strong disturbance to the thermal boundary layers. Also, the induced transverse and longitudinal vortices would dramatically enhance heat exchange inside the thermal boundary layers. Typically, a high Nu/Nu_0 value of 2.1 is achieved in the backward direction. Re is important to the convective heat transfer for heat sink. Nu/Nu_0 in the backward direction is increased by 22.7 % at $Re = 300$, comparing to forward direction. To show advantages of the as-designed microchannels, Table 2 summarizes the overall performance and mechanisms of heat transfer among different copper microchannel configurations. This table indicates that our design outperforms most of microchannel configurations.

Fig. 20. Variations of PEC with different D_h .Fig. 19. Diodic performance with different D_h .

5.4. Diodic performance

As aforementioned, this novel Tesla-type design exhibits significant difference of thermal and hydrodynamic performances in both flow directions. The diodic performances regarding pressure and thermal diodicities are presented in Fig. 14. Based on Eq. (20), the variation of Di_p against Re was calculated. Actually, the increasing Re would inevitably result in higher Di_p . A value of 1.9 is demonstrated at $Re = 400$. The accumulative effect of periodic Tesla-type bends is attributed to the high flow resistance in the backward direction. Significant flow diodic performance has also been reported in the previous researches [2,12]. Fig. 15 compares the pressure diodicities in the existed microchannel configurations [2,11,12,25,40]. Moreover, thermal diodicity is observed between the forward and backward direction. By enhancing fluid mixing, the Di_t reaches to 1.3 at $Re = 400$ in backward direction.

5.5. Thermal resistance and performance evaluation

The numerical study indicates that thermal performance is significantly increased in this design (see Fig. 13). The enhancement of Nu is up to 2-fold compared to conventional microchannel. A comprehensive comparison is conducted between as-designed microchannel and plain microchannel. Fig. 16 depicts the change of average wall temperature, which is substantially reduced in Tesla-type microchannels. Typically, the average wall temperature in the forward and backward directions are reduced by 10.2 K and 12.0 K, respectively, at $Re = 250$. Furthermore, the higher flow rates would enhance convection, leading to lower wall temperatures [1,18,45]. Moreover, thermal resistance (R_{th}) according to wall temperature is calculated to quantify heat dissipation capacity of as-designed microchannel. R_{th} of this configuration in both directions are significantly smaller than that of plain wall microchannel (see Fig. 17). For example, R_{th} is dramatically decreased by 44.6 % and 49.4 %, respectively, in the forward and backward directions at $Re = 250$.

To evaluate overall performance of this design, performance evaluation criterion (PEC) by considering thermal performance and pumping power was calculated referring to Eq. (19). Heat sinks with high PEC is very desirable for potential applications. In this study, Fig. 18 illustrates that forward PEC value is slightly higher than that in the backward direction owing to significantly lower pressure drop, the PEC is increased by 6 % at $Re = 200$. Backward pressure drop is about 2-fold higher than that in the forward direction. Significant difference of overall performance in the different flow directions was reported by the existed studies [25,40].

5.6. The effect of D_h on heat transfer

Hydrodynamic and thermal performances are significantly influenced by the dimensions of microchannel. Overall performance with different D_h was investigated at the same inlet flow rate, the microchannel with D_h of 0.4 mm is selected as the base case. In the Fig. 19, it is found that microchannels with smaller D_h exhibit higher Di_p . This is attributed to the strong fluid mixing in a smaller channel. Particularly, in the backward direction, the fluid flow experienced strong disturbance, accompanying with increased flow resistance [25,34]. The Di_p are 1.93, 1.80 and 1.61, respectively, for microchannel with D_h of 0.3, 0.4 and 0.5 mm at $Re = 350$. The relationship between PEC and different D_h is described in Fig. 20. At lower flow rates, the difference of PEC between different D_h is very small. As inlet Re increases to 200, flow resistance in microchannel with smaller D_h dramatically increases, resulting in lower PEC.

6. Conclusions

This work has conducted a comparative study to elucidate the enhanced mechanism of this as-designed Tesla-type microchannel

configuration. Significant enhancement of thermal performance is achieved through periodic interruption of liquid flow boundary layer and thermal boundary layer as well as strong mixing. Hydrodynamic and thermal performance have been comprehensively investigated through experimental and numerical studies. Nu is much higher than that of conventional plain microchannels.

The periodic Tesla-type bends of this novel design is helpful to heat transfer enhancement due to continuous interruption of thermal boundary layer, sufficient mixing and chaotic advection. Compared to conventional plain-wall microchannel, the average wall temperature is significantly reduced by about 8.4 K to 16.8 K for different working conditions. Consequently, the thermal resistance is decreased by 28 % to 48 %. Furthermore, the PEC of this design is ranging from 1.1 to 1.3 in the forward direction. For $D_h = 0.5$ mm, a PEC of 1.3 is achieved. This microchannel configuration also exhibits significant flow and thermal diodicities. The Di_p and Di_t can be reached to 1.9 and 1.3, respectively, for microchannel configuration with $D_h = 0.4$ mm. Finally, we believe this novel design would provide insightful understandings for the design of effective heat sinks in practical applications.

CRediT authorship contribution statement

Qun Han: Conceptualization, Validation, Writing. **Wenhao Lai:** Validation. **Zhaoxuan Liu:** Validation. **Lirong Li:** Funding acquisition, Methodology, Writing – review & editing. **Wenming Li:** Conceptualization, Funding acquisition, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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